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12.11 The normal shock wave

A shock wave involves the transition from a uniform upstream flow to a uniform downstream flow. In the case of a normal wave, there is no stream speed in the plane of the wave, and the transition is from a supersonic flow to a subsonic flow. The *shock Mach number* $(Ma)_S$ is defined as the ratio of the speed of the wave, relative to the upstream gas, to the speed of sound in this gas. The shock wave is stationary in the steady flow frame of reference, and the shock Mach number may then be identified with the Mach number of the upstream flow. The wave is generally regarded as a discontinuity in a continuum flow and the ratios of the stream speed, density, pressure, and temperature across the wave are given by the Rankine–Hugoniot relations (Liepmann and Roshko 1957). In fact, the shock wave has a finite thickness and the profile depends on the transport properties of the gas.

The Navier–Stokes equations may be applied to the shock structure problem and yield an ordinary differential equation that is amenable to numerical solution (Gilbarg and Paolucci 1953). A shock wave thickness L may be defined as the distance required to span the overall density change by the maximum density gradient within the wave. This thickness is generally expressed as a multiple of the upstream mean free path λ_1 and is a function of the viscosity–temperature index ω and $(Ma)_S$.

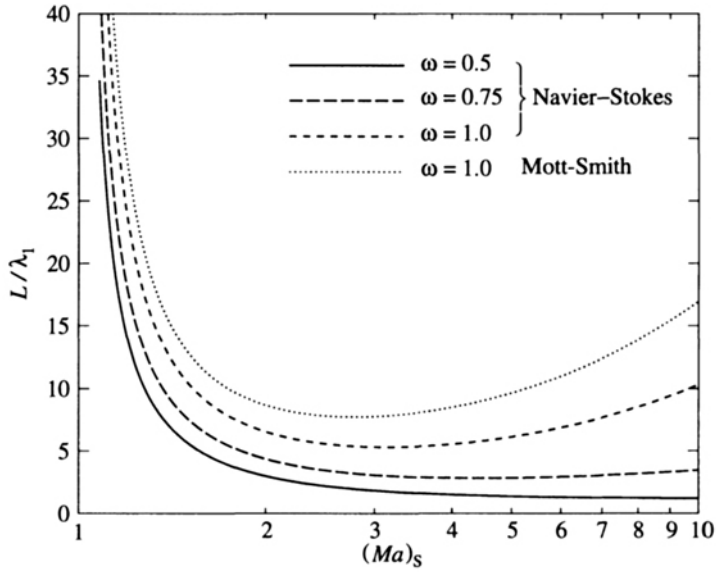


Fig. 12.23 Theoretical shock wave thickness in a monatomic gas.

The predictions of the Navier–Stokes theory for a monatomic gas are shown in Fig. 12.23. These employ the definition of eqn (4.52) for the mean free path and this means that the parameter 'β' in the analysis of Gilbarg and Paolucci becomes

$$\frac{(30\pi)^{1/2}(\alpha+1)(\alpha+2)}{3\alpha(5-2\omega)(7-2\omega)}.$$

For $\alpha = 1$ and $\omega = \frac{1}{2}$, this reduces to the hard sphere value that has been used in most applications of the theory. The calculations that led to Fig. 12.23 set the VSS scattering parameter α to unity. It is the change in the definition of mean free path that is largely responsible for the significant effect of ω on the width of very weak waves that is apparent in Fig. 12.23. The ratio of the 'Maxwell' value to the 'hard sphere' value of L/λ_1 becomes even larger for very strong waves in that the former exhibits a minimum of approximately five at a shock Mach number near three and then increases, while the latter tends to a value near unity. The maximum slope thickness of a gas with a realistic value of ω has a shallow minimum of two to three upstream mean free paths. These thin waves involve such large gradients of the macroscopic gas properties that the Chapman–Enskog theory ceases to be valid and the Navier–Stokes equations fail. Figure 12.23 also shows the result from the Mott-Smith theory of §8.2.

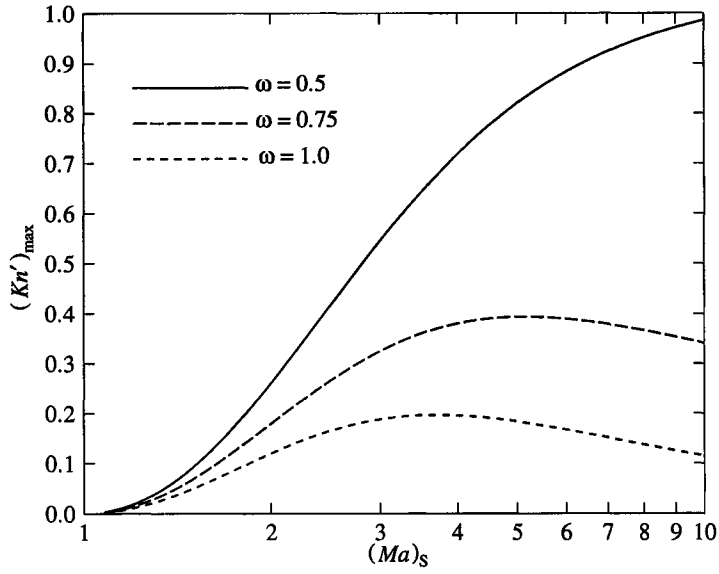


Fig. 12.24 The maximum value of the local Knudsen number in the Navier–Stokes shock profiles.

The onset of significant error in the Chapman–Enskog theory and the Navier–Stokes equations was linked in §1.2 to a value of 0.1 for the local Knudsen number based on eqns (1.1) and (1.2). The maximum values of this local Knudsen number in the Navier–Stokes shock wave profiles was therefore calculated and the results are plotted in Fig. 12.24. This indicates that, for a realistic value of ω, the error in the Navier–Stokes profile becomes significant at a shock Mach number of about 1.6, and very serious at shock Mach numbers greater than two.

High Knudsen numbers that are generally associated with rarefied gas flows therefore occur in strong shock waves at all gas densities. The waves in many flow configurations are well away from the influence of solid surfaces and are not subject to the uncertainties associated with gas–surface interactions. There is a considerable body of well-substantiated experimental data for the structure of normal shock waves, and this flow has provided the most important test case for the verification of numerical methods in molecular gas dynamics.

A number of qualitatively different boundary conditions have been employed in DSMC studies of normal shock waves. One of these could be applied to the general one-dimensional program DSMC1.FOR. This would employ the 'type 3' or stream option at each boundary. The properties of one stream would correspond to the uniform conditions upstream of the wave, while those of the other would correspond to the conditions downstream of the wave. The only modification would be to set half the gas in the initial state to the upstream conditions and the other half to the downstream conditions. There would then be an initial discontinuity or step at the desired location of the centre of mass of the wave. The steady shock profile would develop during the unsteady phase of the calculation. This scheme has led to acceptable results for strong shock waves as long as the number of simulated molecules is large and the sample is relatively small. The problem is that the number of molecules entering the upstream and downstream boundaries fluctuates and the total number of molecules is subject to a random walk effect. Since there is nothing to hold the shock in place, its location is also subject to a random walk and this causes a smearing of the time-averaged profile.

The subsonic downstream boundary at x_d causes the greater part of the problem and it may be replaced by a specularly reflecting solid surface that moves with speed u_s equal to the downstream Rankine–Hugoniot speed. Then, for molecules that move to x beyond x_d, the post-reflection value of the velocity component u is

$$u' = 2u_w - u. \tag{12.30}$$

The post-reflection location of the molecule is readily shown to be

$$x' = 2x_w - x + u'\,dt, \tag{12.31}$$

where dt is the time step. Should x' remain beyond x_d, the molecule is removed. The boundary condition is essentially a moving piston that, after interacting with the molecule, 'jumps' back to its original position. It has

been implemented in program DSMC1S which is an adoption of the general one-dimensional program DSMC1. This boundary condition has the further advantage that it requires only the downstream speed to be calculated from the Rankine–Hugoniot theory, while the stream boundary condition also employs the theoretical values of the density and temperature. It essentially amounts to the production of a shock wave by a moving piston with the Rankine–Hugoniot velocity ratio used to transform the frame of reference so that the wave is stationary.

The other changes to DSMC1 include the removal of the boundary options that are no longer required, and the tailoring of the input and output to the shock wave problem. In addition, the initial state of the gas is a discontinuous wave at the origin. Even with the piston boundary condition, the total number of molecules is subject to random walk and the time-averaged shock profile can be smeared by the consequent movement of the wave. The subroutine STABIL has been added to prevent this. Should the total number of molecules depart from the initial number by more than a small specified number, all molecules are displaced by the same distance. The molecules that move across one boundary are removed and those that are initially within the displacement distance from the other boundary are duplicated. The displacement distance is calculated on the basis of the upstream and downstream Rankine–Hugoniot densities as that required to restore the number of molecules to the original value.

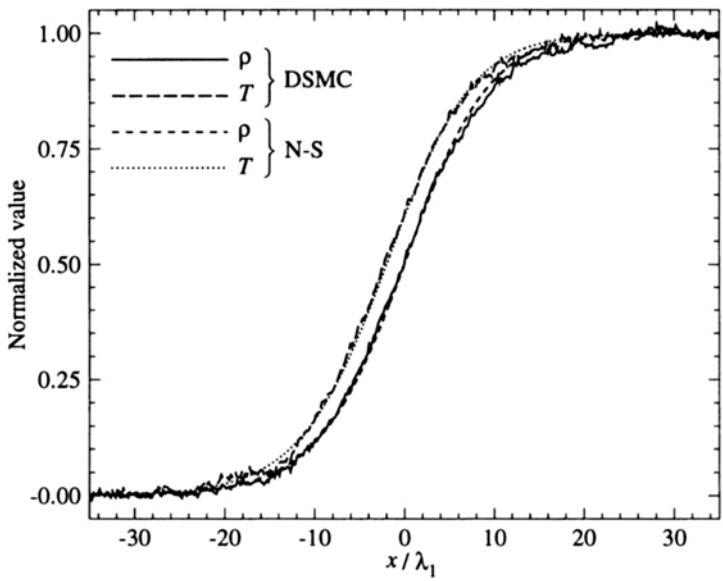


Fig. 12.25 Temperature and density profiles in argon with $(Ma)_S = 1.2$.

A wave with a shock Mach number of 1.2 in argon at 293 K has been chosen as the first test case. The VSS molecular model was used with parameters for argon that are listed in Appendix A. The DSMC calculation employed 27,000 simulated molecules in 400 cells, each divided into six sub-cells. The mean separation between collision pairs was less than one hundredth of the upstream mean free path, and the time step was just over one twentieth of the mean time between collisions. The computational requirements of the DSMC method were therefore satisfied by a very wide margin. Steady flow was assumed after the flow had moved approximately 20 shock widths and the time average was taken over the time for the flow to move several thousand shock widths. This involved the computation of some 43,000,000 collision events.

The results for the density and temperature profiles are compared in Fig. 12.25 with the predictions of the Navier–Stokes equations. The normalized density is defined by

$$\hat{\rho} = \frac{\rho - \rho_1}{\rho_2 - \rho_1}, \tag{12.32}$$

where the subscripts 1 and 2 represent the upstream and downstream Rankine–Hugoniot values, respectively. The normalized values of the other macroscopic flow variables are defined in a similar manner. As expected, the two sets of results are in good agreement for this very weak wave.

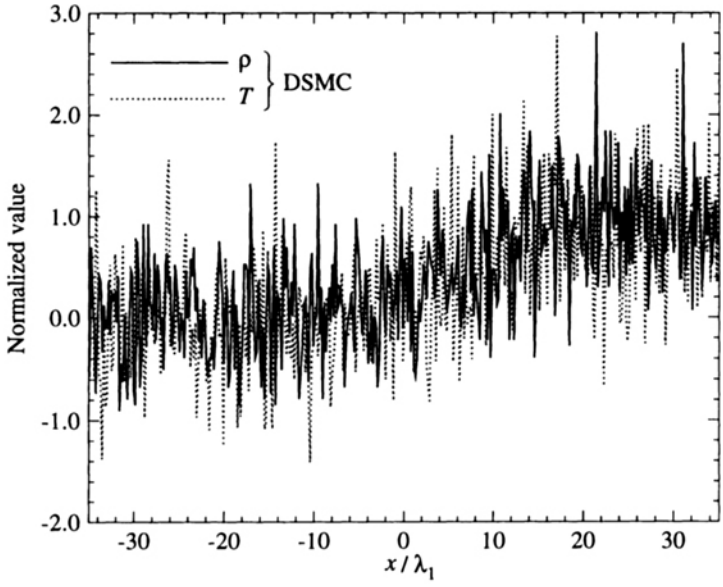


Fig. 12.26 A typical profile based on instantaneous values.

This is a demanding calculation in that it involves relatively small disturbances with the density ratio being only 1.2973. Since the statistical scatter is a fraction of the density, it is a larger fraction of the overall disturbance than is the case with strong shock waves. This is illustrated in Fig. 12.26 which shows the temperature and density profiles from a typical instantaneous sample of the flow. The number of molecules per cell is such that the overall density change across the wave corresponds to about two standard deviations of the statistical scatter. This means that there are a number of cells in which the 'noise' exceeds the 'signal'.

The importance of the stabilization of the shock location through the control of the fluctuation in the total number of molecules in the subroutine STABIL was tested by repeating the calculation with this subroutine disabled. The number of molecules started at 27,022 and executed a random walk. The displacement was generally downward and its magnitude at larger times was typified by the final value of 26,400. This led to a general rearward displacement of the wave and, as shown in Fig. 12.27, the consequent distortion of the time-averaged profiles is very serious. While the net shock displacement was downstream in this case, it could equally likely have been in the upstream direction. Fortunately, the need for direct intervention to control random walks is unique to this flow because the flow configuration is almost always fixed by the geometry of the physical boundaries.

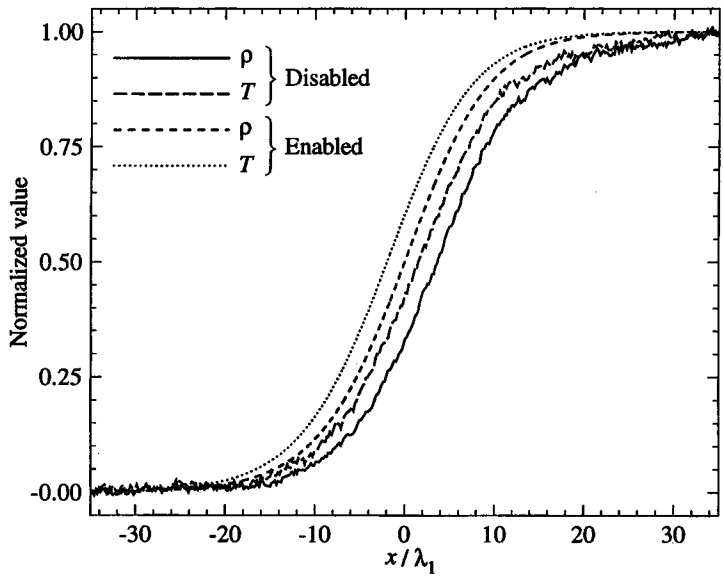


Fig. 12.27 The effect of disabling the subroutine STABIL.

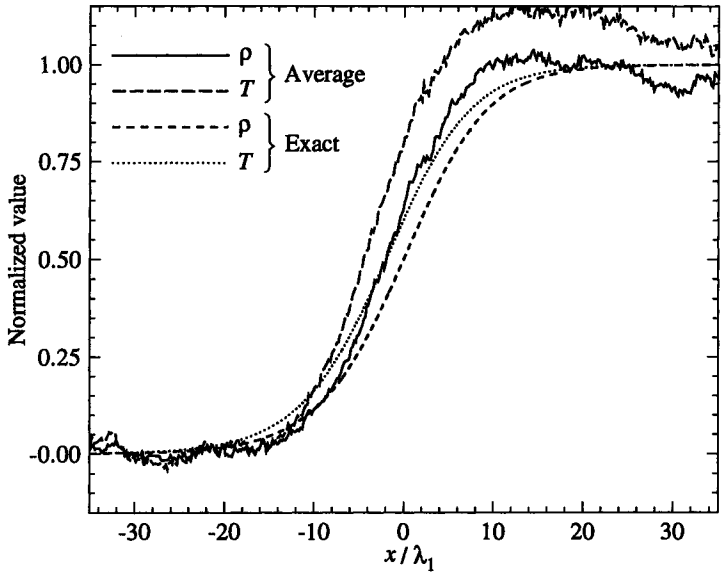


Fig. 12.28 The effect of average rather than exact conservation of momentum and energy in collisions.

As noted in §10.2, some versions of the DSMC method attempt to make the procedures more consistent with the Boltzmann equation by replacing the velocity components of only one of the two molecules in each collision. It was pointed out in §10.4 that, while this conserves momentum and energy on the average, they are not conserved in each collision and this introduces a random walk in the overall momentum and energy of the gas. The program DSMC1S was modified so that the number of collisions was doubled, but only one of the molecules was changed in each collision. If there were no random walks, the results would not be affected. However, as shown in Fig. 12.28, the solution is degraded almost to the point of being rendered useless. It is ironic that a mathematical 'convergence proof' has been presented (Babovsky and Illner 1989) for the procedures that include the random walks, but not for the regular procedures. The decline in the statistical scatter with the growth of the time-averaged sample that gradually transforms Fig. 12.26 to Fig. 12.25 is sometimes referred to as 'convergence'. This is undesirable because the statistical process is quite unrelated to the mathematical concept of convergence. The mathematical model of the simulation did not take into account the possible random walk effects, and these are the real determinant of whether or not an accurate solution is obtained from a probabilistic physical simulation. Note that the velocities of both molecules are changed in the collisions in real gases and, from the physical point of view, the modification is nonsense.

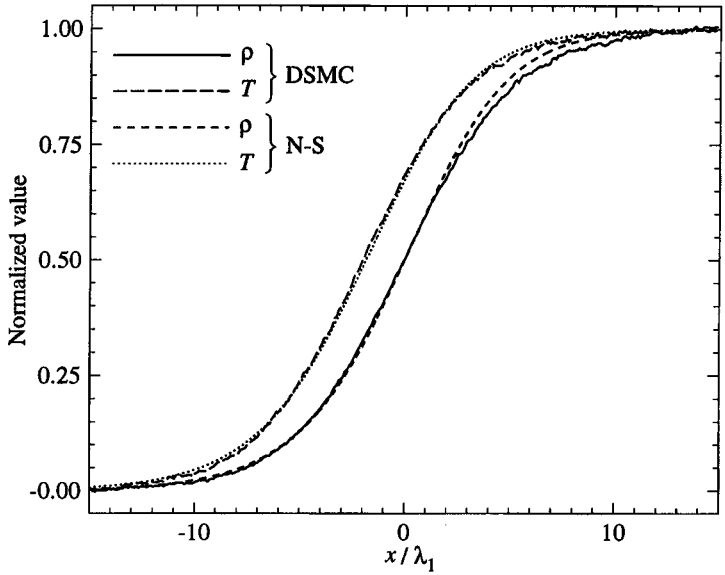


Fig. 12.29 Temperature and density profiles in argon with $(Ma)_S = 1.4$.

The data in the listing of program DSMC1S is for a shock Mach number of 1.4 in argon at 293 K. The density and temperature profiles for this case are shown in Fig. 12.29. This shows the expected reduction in thickness and there is also a significant increase in the separation of the temperature and density profiles. The scatter is reduced as a result of the larger disturbance and, although the shock Mach number is still well within the expected limit on the validity of the Navier–Stokes equations, there is a significant discrepancy in the density profiles for the downstream half of the shock wave. This difference has been observed consistently in studies of weak shock waves and Phan-Van-Diep et al. (1991) showed that it is consistent with the predictions of Simon and Foch (1977) for the differences that are introduced when the Navier–Stokes equations are replaced by the Burnett equations. The maximum slope thickness is clearly inadequate for the quantification of this difference and recourse must be had to the *shape factor* Q_ρ that was defined by Schmidt (1969) as

$$Q_\rho = \int_{-\infty}^{0} \hat{\rho}\, dx \Big/ \int_{0}^{\infty} (1-\hat{\rho})\, dx, \tag{12.33}$$

where the origin is at $\hat{\rho} = 0.5$. There is a qualitative difference between the prediction of the Navier–Stokes equation that Q_ρ should increase monotonically with shock Mach number and that of the Burnett equations

that Q_ρ should decrease to a minimum near $(Ma)_S = 1.3$. The Navier–Stokes value of Q_ρ for $(Ma)_S = 1.4$ is 1.09 and the Burnett value of Simon and Foch is 0.95. These values are for $\omega = 1$, and the calculations of Erwin, Muntz, and Pham-Van-Diep indicate that the expected values for a realistic ω would be about 1% lower. The DSMC density profile of Fig. 12.29 is in good agreement with the Burnett value with an estimated maximum error band of ±5%. The Burnett equations also predict a separation between the density and temperature profiles at the midpoint of the wave that is about 10% higher than the Navier-Stokes value. The DSMC result is consistent with this, but the difference is of the order of the error band for this quantity.

The corresponding results for a shock Mach number of two are shown in Fig. 12.30. The local Knudsen numbers of Fig. 12.24 indicate that there should be clear errors in the Navier–Stokes profiles for this case, and the DSMC results confirm this. The maximum slope thickness from the Navier–Stokes theory is about 25% smaller than the value of $L/\lambda_1 = 6.2$ from the DSMC calculation. This may appear to be in conflict with the value of 4.8 that was obtained from the experimental study of Alsmeyer (1976). However, the mean free path that is used here is, for $\omega = 0.81$ and $\alpha = 1.6625$, equal to 76% of the hard sphere mean free path that was used by Alsmeyer. Therefore, in terms of the actual thickness, the two results agree to within one or two percent.

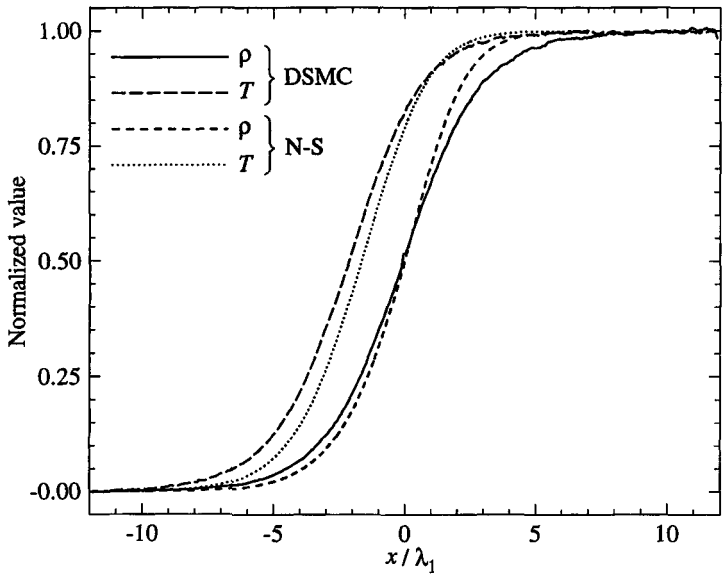


Fig. 12.30 Temperature and density profiles in argon with $(Ma)_S = 2$.

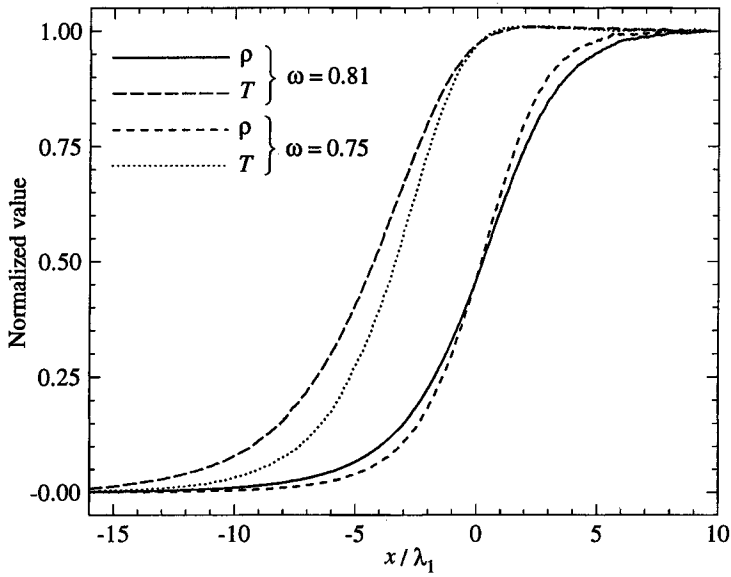


Fig. 12.31 The structure of a strong shock of $(Ms)_S = 8$ in argon.

According to the Burnett theory, the shape factor Q_ρ should be about unity for $(Ma)_S = 2$, but the value from the integration of the profile in Fig. 12.30 remained at the value of 0.95 that was obtained for the $(Ma)_S = 1.4$ case. This is at the limit of the estimated error bound and the result is inconclusive. On the other hand, the separation between the temperature and density profiles at the higher Mach number now supports the Burnett theory to well within the expected scatter. The Maxwell gas result of Simon and Foch predicts 30% more separation than that given by the Navier–Stokes theory. The separations are less for the more realistic molecular model that is used here, but the percentage increase is consistent with the predicted value.

There are qualitative changes to the profile as the shock becomes stronger. Figure 12.31 is for $(Ma)_S = 8$ and this shows a marked asymmetry in the profiles and also an overshoot of about 1% in the temperature. The asymmetry is accentuated by a region at the front of the wave where the density is essentially equal to the upstream value, but where there is a significant increase in temperature. The relative velocity between the unaffected stream molecules and any affected molecules is so large that a minute proportion of the latter have a noticeable effect on the temperature. The velocity distribution is qualitatively bimodal, but is not the simple mixture of the upstream and downstream Maxwellians that was assumed by Mott-Smith. This has been confirmed by Phan-Van-Diep et al. (1989) who compared the DSMC and Mott-Smith distributions with measured

distributions in helium at $(Ma)_S = 25$. The distributions from the DSMC calculations were in excellent agreement with the experimental values. The distribution near the front of the wave was strongly bimodal, but the Mott-Smith theory provided a poor representation of the merging of these distributions.

The maximum slope thickness for the value of 0.81 for ω that is appropriate at temperatures near the standard temperature is significantly thicker than Alsmeyer's experimental value. Figure 12.31 also shows the profiles for $\omega = 0.75$. When allowance is made for the differing definitions of the mean free path, the thickness is in good agreement with the measured value.

If judged by the results for the shock wave, it might be thought that the use of the VHS or VSS mean free path of eqn (4.52) in place of the hard sphere mean free path of eqn (4.56) is disadvantageous. This is definitely the case for extremely weak waves because it leads to a dependence on ω of the normalized thickness, while both the actual thickness and the thickness based on the upstream hard sphere mean free path do not depend on ω. Furthermore, the profiles for the very strong shock also appear to depend on ω. However, the Knudsen numbers based on the mean free path of eqn (4.52) should employ, in this equation, the collision cross-section at the relative velocity that is most significant for the problem. In the case of a strong shock wave, the most significant collisions occur with a relative velocity of the order of the upstream velocity, and this corresponds to that at a temperature of the order of the downstream temperature. This means that the profiles of strong shock waves should be independent of the molecular model if they are plotted against distance normalized to the downstream rather than the upstream mean free path. The downstream density is a constant in this change, and the difference comes from the change in cross-section. Eqns (3.60) and (3.67) show that the cross-section is inversely proportional to the temperature to the power $\omega - \frac{1}{2}$. Therefore, for the temperature ratio of 20.87 across a $(Ma)_S = 8$ shock in a monatomic gas, the effective ratio of the downstream mean free path at $\omega = 0.75$ to that at $\omega = 0.81$ is 0.833. This would bring the two sets of shock profiles into near coincidence.

Because the velocity difference between the two components of the near bimodal distribution at the front of the wave is along the x-axis, the 'parallel temperature' based on the velocity component in the x-direction is larger than the 'normal temperature' based on the velocity components in the y- and z-directions. This effect is illustrated in Fig 12.32 which shows that extreme nonequilibrium persists over the whole of the region upstream of the centre of the wave, and the gas comes to equilibrium in the downstream half of the wave. The parallel temperature reaches a maximum value that is one third higher than the final downstream temperature. Although these very high values of T_x/T_n are characteristic of strong shocks, this ratio is above unity even in a very weak wave. Its value could be calculated from the Chapman–Enskog distribution functions that accurately describe the transport properties in very weak waves.

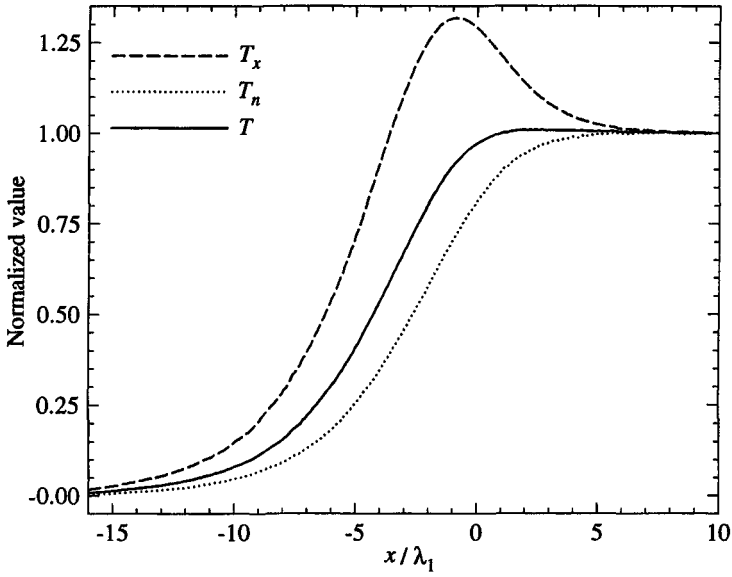


Fig. 12.32 The parallel and normal temperatures in an $(Ma)_S = 8$ shock.

In the case of a diatomic gas with a finite rotational relaxation time, the rotational temperature must lag behind the translational temperature. The program DSMC1S includes the routines that are required for the simulation of molecules with rotational degrees of freedom. A calculation was made for a shock of Mach number 1.71 in nitrogen. The VHS model was used with the parameters for nitrogen that are listed in Appendix A, and the rotational relaxation collision number was set to five. The results for the density and temperature profiles are shown in Fig. 12.33. All the computations in this section are of similar magnitude with regard to CPU time, and the higher level of scatter in calculations for weak shock waves is very noticeable.

The profiles of density and overall temperature are qualitatively similar to those in Figs. 12.25 and 12.29 for weak shocks in a monatomic gas. The significant effects of the internal degrees of freedom relate to the component temperatures and, as expected, the translational temperature profile leads the overall temperature. Also, there is a small overshoot of about one or two percent in the overall temperature. The rotational temperature lags the other temperatures and its profile, in this case, is very close to the density profile. The reason for choosing this case is that measured values of the density and rotational temperature profiles were obtained by Robben and Talbot (1966). They found that the rotational temperature profile lagged slightly behind the density profile, and the DSMC results of Fig. 12.33 are consistent with the experiments.

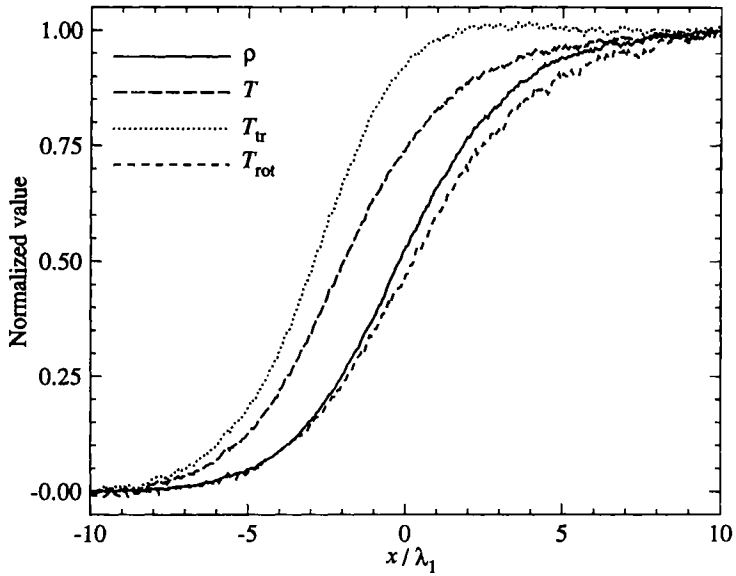


Fig. 12.33 $(Ma)_S = 1.71$ shock in nitrogen with $Z_{rot} = 5$.

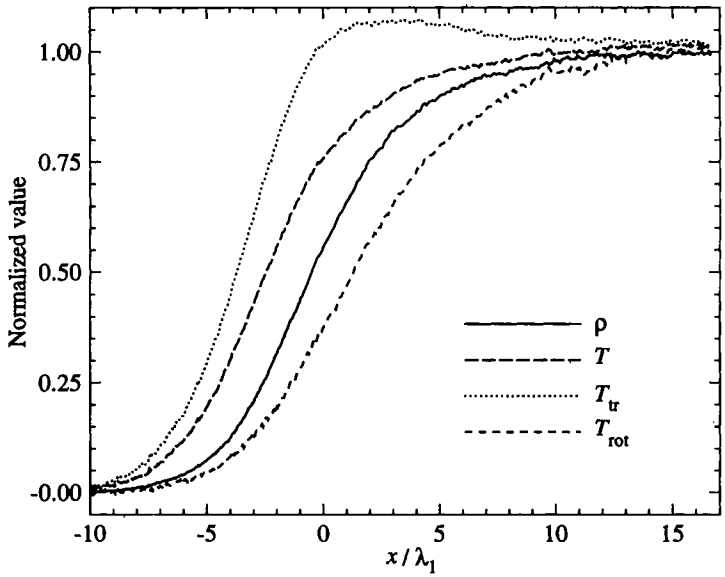


Fig. 12.34 $(Ma)_S = 1.71$ shock in nitrogen with $Z_{rot} = 10$.

The calculation was repeated with a rotational relaxation collision number Z_{rot} of 10, and the resulting profiles are shown in Fig. 12.34. The separation between the translational and rotational increases, but is significantly less than double the spacing in the $Z_{rot} = 5$ case. On the other hand, the overshoot in the translational is more than doubled. The rotational temperature profile lags significantly behind the density profile, and the experiments of Robben and Talbot are indicative of a rotational relaxation collision number of approximately five for nitrogen. An interesting and perhaps unexpected result is that the shock wave thickness based on the maximum density slope is directly affected by Z_{rot}. In this case, it increased by approximately 20% as Z_{rot} changed from 5 to 10.

Program DSMC1S.FOR can also be applied to gas mixtures. The most interesting experimental results for shock waves in gas mixtures are those of Gmurczyk et al. (1979) for helium with a small proportion of Xenon. These exhibited the leading of the heavy gas profile by the light gas profile that had long been known from experimental and numerical studies. The novel points were a pronounced and relatively sudden change in the slope of the helium profile and an overshoot in that profile. The former has been confirmed in a number of DSMC studies, but the latter appears to be an artefact of the measurement technique. These findings have been illustrated through a calculation with DSMC1S for a shock of Mach number 3.89 in a mixture of 3% xenon in helium.

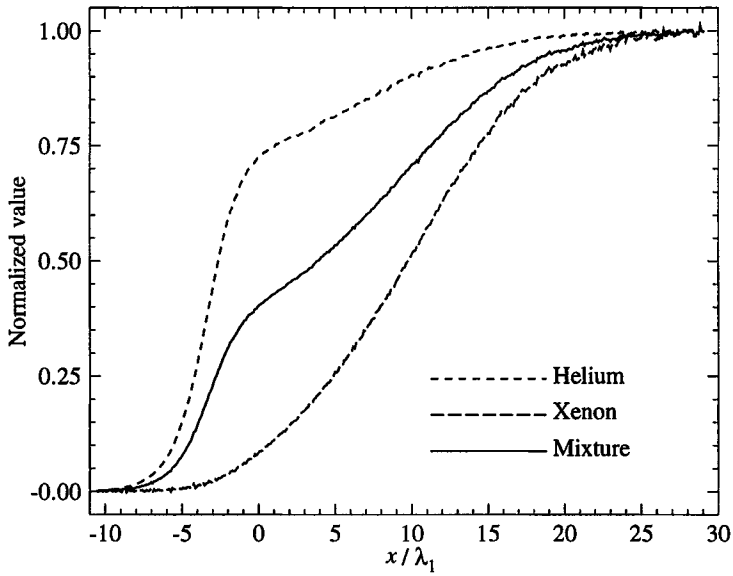


Fig. 12.35 Density profiles in the helium–xenon mixture.

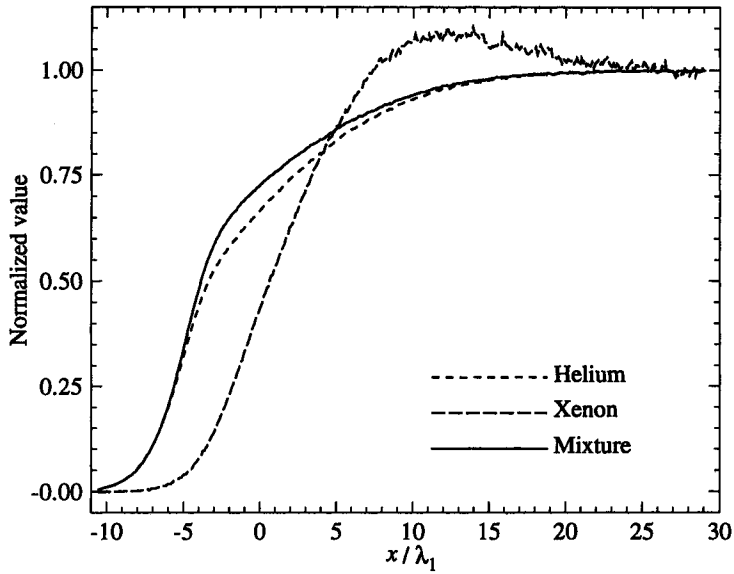


Fig. 12.36 The temperature profiles in the helium–xenon mixture.

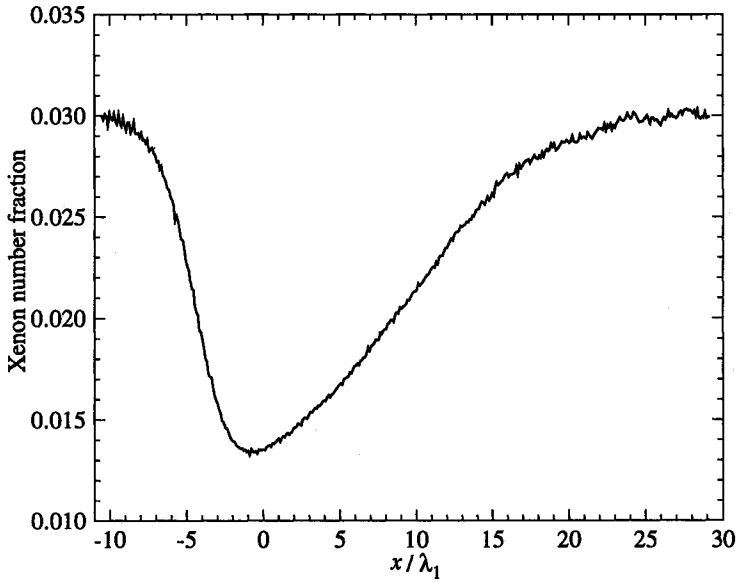


Fig. 12.37 The xenon concentration in the shock wave.

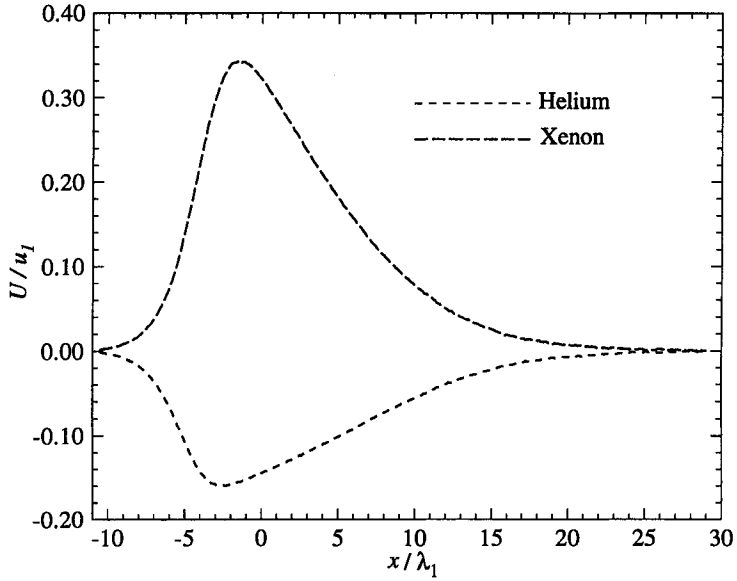


Fig. 12.38 The diffusion velocities in the shock wave.

The VSS model was used with the option that allows separate values of ω and α for all combinations of molecules. The density and temperature profiles are shown in Figs 12.35 and 12.36, respectively. The component densities, like the densities in the simple gas shocks, can be regarded as either number densities or mass densities, while that for the mixture must be a mass density. The origin is at the 'number centre' of the wave. The helium profile exhibits the expected sudden change in slope and there is a suggestion that this is followed by a point of inflection. There is a clear point of inflection in the mixture profile, and this can be said to exhibit a 'bump'. Because of equipartition, the overall temperature is very close to the helium temperature. The temperature profiles lead the density profiles, and the xenon temperature has an overshoot of about 10%. Since the upstream speed would correspond to a shock Mach number of 15.93 in pure xenon, this overshoot is surprisingly small.

Note that all figures that show DSMC results are plotted directly from the output, with the sampled values shown either as points or as line segments between these points. No form of smoothing has been applied to any of the results. The scatter is, of course, much higher in the xenon which is present in a small concentration. Also, as shown in Fig. 12.37, the xenon concentration at the centre of the shock drops to less than half the upstream and downstream values. The diffusion velocities that are associated with this separation effect are shown in Fig. 12.38 as a ratio to the upstream velocity.

The Mott-Smith moment method for the structure of strong shock waves was presented in §8.2 and a solution was obtained for the case of Maxwell molecules. Program DSMC1S has been used to calculate the structure of a shock of Mach number 8 in a monatomic simple gas of Maxwell molecules. The normalized number density profile from the DSMC calculation is compared in Fig. 12.39 with the Mott-Smith profile calculated directly from eqn (8.39). The maximum slope thicknesses based on these profiles are almost identical, but there are significant differences in the structure. The location of the origin is such that

$$\int_{-\infty}^{0}(n-n_1)\,dx = \int_{0}^{\infty}(n_2-n)\,dx\,,$$

and, for the symmetrical Mott-Smith profile with $Q_f = 1$, the origin is at a normalized number density of 0.5. The actual profile has a shape factor Q_ρ well in excess of unity, and this is characteristic of strong shock waves.

Program DSMC1S samples in array SDF the information on the axially symmetric velocity distribution function in a set number of cells. The axial and radial dimensions of the axially symmetric velocity space are divided into 400 and 200 intervals, respectively, and the number of molecules in each of the resulting velocity space cells are output in file DSMCDF.OUT. This defines the velocity distribution function, and this may be partially integrated to give the axial function f_x and the radial or normal function f_n.

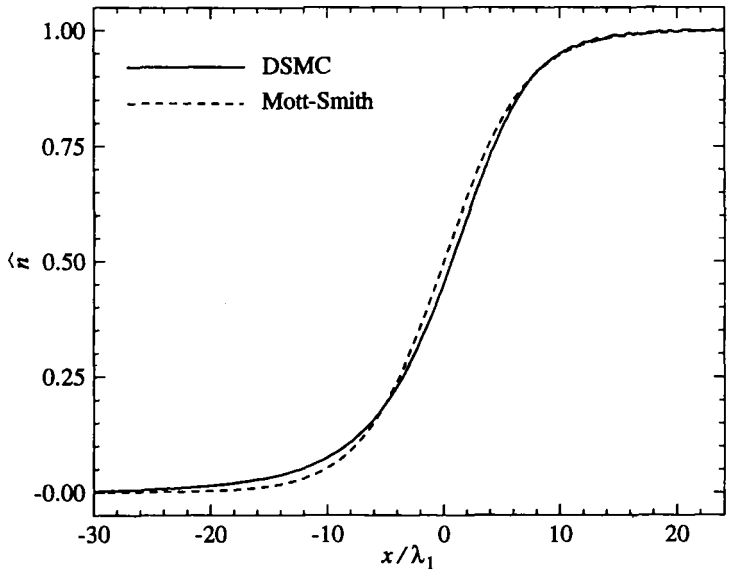


Fig. 12.39 The normalized number density profile for a shock of $(Ma)_S = 8$ in a simple gas of Maxwell molecules.

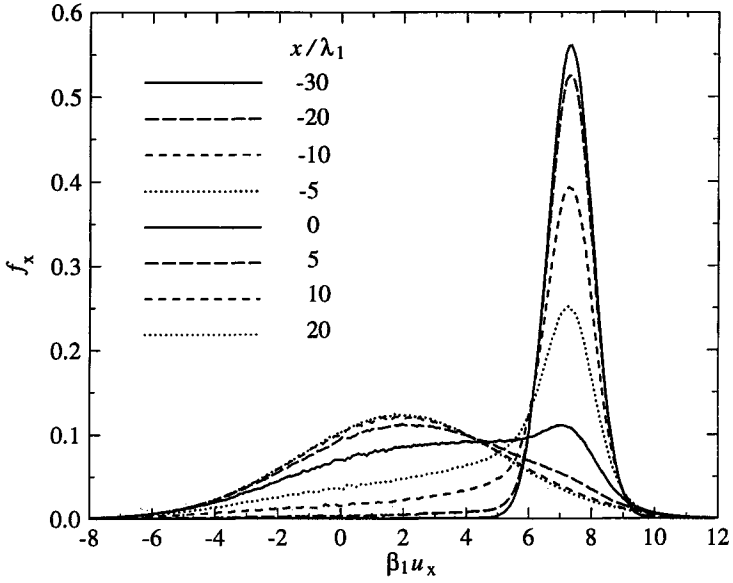


Fig. 12.40 DSMC result for the axial distribution function.

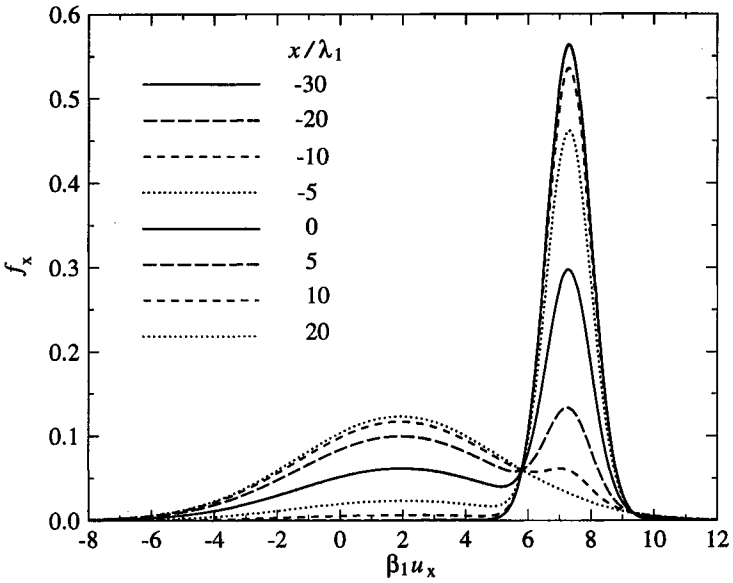


Fig. 12.41 The Mott-Smith prediction of the axial distribution function.

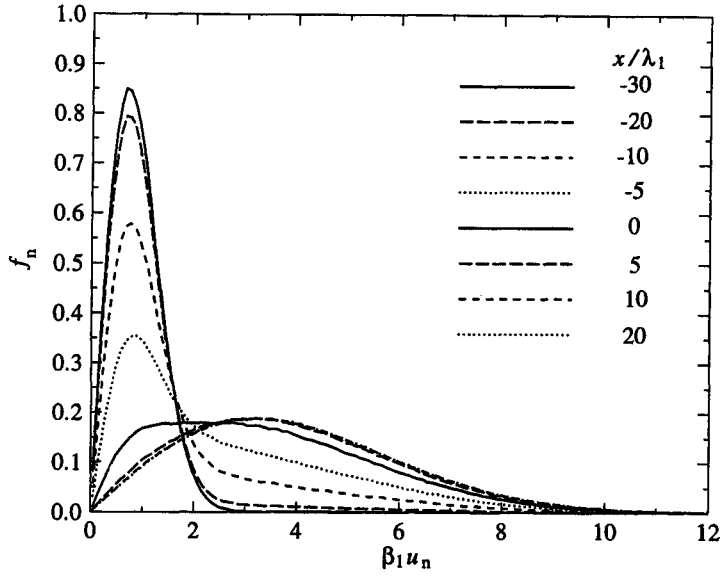


Fig. 12.42 DSMC result for the radial distribution function.

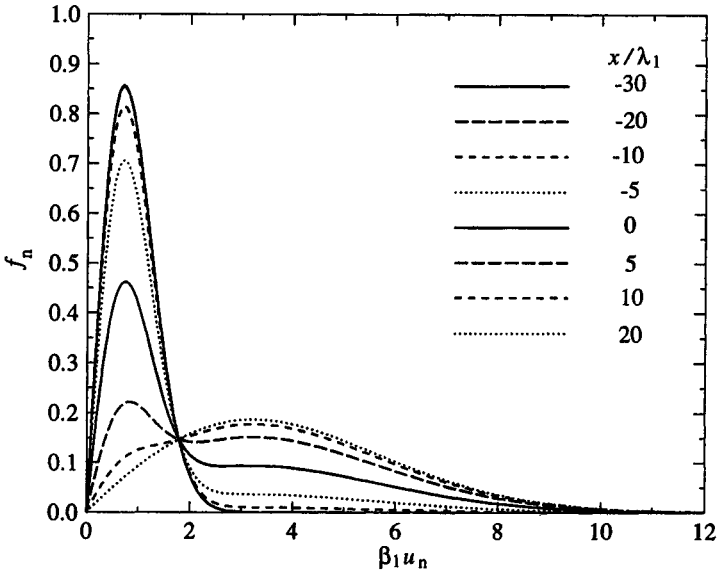


Fig. 12.43 The Mott-Smith prediction of the radial distribution function.

The shortcomings of the Mott-Smith theory are shown most clearly in the details of the velocity distribution functions. The DSMC and Mott-Smith profiles for the distribution function of the axial component of velocity are plotted in Figs 12.40 and 12.41, respectively. The corresponding results for the radial component of velocity are shown in Figs 12.42 and 12.43. The Mott-Smith model overestimates the degree to which the upstream distribution persists to the centre of the wave. This leads to a distribution function with a double peak at the centre of the wave. The actual distribution does not show a double maximum at any location, but has a point of inflection at the centre of the wave. The differences in the radial distribution function are similar, but there is not even a point of inflection in the DSMC profile at the centre of the wave.

The most basic information on the flow is given by the full distribution function and the DSMC result at the centre of the wave is shown in Fig. 12.44. This is defined such that the fraction of molecules with axial component between $\beta_1 u_x$ and $\beta_1 u_x + \mathrm{d}(\beta_1 u_x)$ and radial component between $\beta_1 u_n$ and $\beta_1 u_n + \mathrm{d}(\beta_1 u_n)$ is equal to $f\mathrm{d}(\beta_1 u_x)\,\mathrm{d}(\beta_1 u_n)$. As in the partially integrated distributions, there is a peak in the distribution at the most probable velocity in the upstream gas, but no peak at the downstream most probable velocity. The values in critical regions of the corresponding Mott-Smith distribution would be in error by factors of two or three.

The distribution functions have some resemblance to those for the bimodal distribution but, while the Mott-Smith model is qualitatively useful at the macroscopic level, it is inadequate at the microscopic level. It should also be remembered that the good prediction of the shock thickness by the Mott-Smith theory is dependent on the essentially empirical choice of the macroscopic quantity in the fourth moment equation.

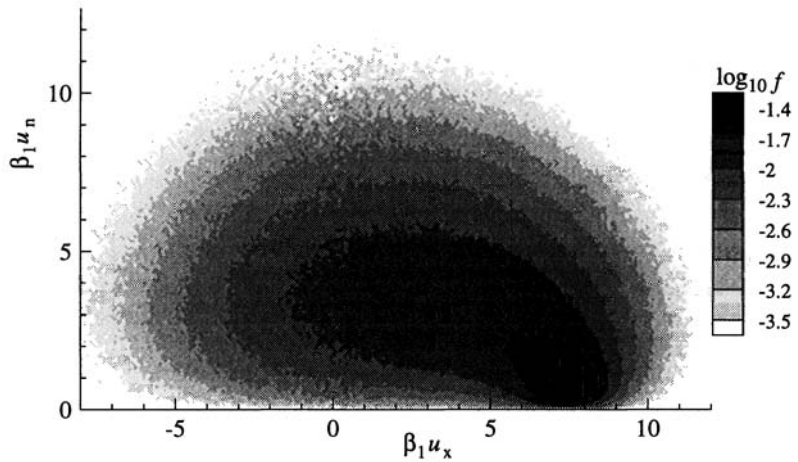


Fig. 12.44 The DSMC distribution function at $x/\lambda_1 = 0$.

12.12 Stagnation streamline flow

Program DSMC1S for the shock wave structure cannot be applied unless the velocity downstream of the wave is known. This is a serious problem for very strong shock waves in diatomic gases that lead to dissociation, because, while the equilibrium downstream conditions can be calculated, the recombination rate can be so slow in a rarefied gas that these conditions are not attained in any distance that can be calculated. The solution is to calculate the one-dimensional equivalent of the streamline in a two- or three-dimensional flow that reaches the stagnation point.

Program DSMC1T.FOR models the stagnation streamline as a constant area flow with a specified supersonic stream entering at one boundary and a diffusely reflecting surface at the other boundary. Initially, the gas has the properties of the undisturbed stream and an unsteady shock wave propagates from the surface into the flow. At some stage, molecules are removed from the sides of a region that extends from the wall to the specified boundary of this 'removal region' which must be downstream of the shock. The removal is such that the outlet flux is equal to the inlet flux and the flow settles down to a steady state with a stationary shock. The problem is to determine the selection rules such that mass, momentum and energy are conserved along the flow. The following analysis follows that of Bird (1986).

The mass conservation must be enforced on the basis of the number flux of each type of atom that comprise the molecules, because the composition of the gas may change across the shock. The streamline is along the x-axis and, if the rate of removal of molecular mass is $\dot{m}$ per unit volume, the equation for the conservation of mass along the streamline is

$$\mathrm{d}(\rho u) = -\dot{m}\,\mathrm{d}x. \tag{12.34}$$

Then, if $\bar{u}$ is the mean value of the streamwise velocity components of the removed molecules, the equation of conservation of momentum is

$$\mathrm{d}p = -\mathrm{d}(\rho u^2) - \bar{u}\dot{m}\,\mathrm{d}x. \tag{12.35}$$

This can be written,

$$\mathrm{d}p = -\rho u\,\mathrm{d}u - u\,\mathrm{d}(\rho u) - \bar{u}\dot{m}\,\mathrm{d}x$$

and eqn (12.34) shows that the final two terms cancel as long as $\bar{u}$ is equal to u. This means that the continuum momentum equation is satisfied as long as the selection rules for molecular removal are independent of the streamwise velocity component of the molecules.

Finally, the energy conservation equation is

$$\mathrm{d}(\rho u h_0) = \dot{m}\bar{e}_0\,\mathrm{d}x, \tag{12.36}$$

where h_0 is the stagnation enthalpy and $\bar{e}_0$ is the mean stagnation energy of the removed molecules. This equation can be written

$$\rho u\,\mathrm{d}h_0 + h_0\mathrm{d}(\rho u) = -\dot{m}\bar{e}_0\,\mathrm{d}x,$$

and, using eqn (12.34), this reduces to the continuum energy equation

$$dh_0 = 0$$

if the molecules are selected such that

$$\bar{e}_0 = h_0 . \tag{12.37}$$

Eqn (12.37) shows that the average specific energy of the removed molecules must exceed that averaged over all molecules in the flow by RT. This corresponds to an average energy margin of kT over the average translational value of $3kT/2$. Since the selection must be independent of the streamwise velocity component, the energy increment is best achieved by removing the molecules with probability proportional to the jth power of their velocity component normal to the stream direction. It may be shown that the mean translational energy of the removed molecules in an equilibrium gas is then

$$\bar{e}_{tr} = (3+j)kT/2. \tag{12.38}$$

For j=0, this gives the expected value for a stationary gas, and j=1 applies to the flux to or across a surface element, as discussed in §4.2. The requirement of eqn (12.37) is satisfied exactly by $j = 2$, so that the molecules are removed with probability proportional to the square of the velocity component normal to the stream. It can be shown that eqn (12.38) applies to both removal in one plane, as in a two-dimensional flow, or in the radial direction in an axially symmetric flow.

The distribution of the removed molecules along the streamline must also be specified. If the shock structure is the item of interest, the only requirement on the removal region is that it should be well downstream of the shock. However, if the removal starts at the centre of the shock and is otherwise independent of location, there is evidence (Bird 1986) that the overall flow is a good representation of the stagnation streamline in a real axially symmetric flow. The acceptance–rejection technique is used to impose the 'normal velocity squared' removal rule by comparing this quantity with a maximum value. It is uncertain whether a separate maximum should be kept for each species in a gas mixture. This would affect the species separation effects and this program may not correctly model the pressure diffusion that occurs in a real multi-dimensional flow. Program DSMC1T therefore employs separate maximum values for each species and this point will be examined further in Chapter 14.

Program DSMC1T includes the routines from program DSMC0D that deal with vibrational excitation, dissociation and recombination. The routines for the implementation of the stream and surface boundaries come from program DSMC1. The molecule removal procedures that employs the above logic to model a stagnation streamline is applied in the subroutine STAGR. This is the only subroutine that is unique to this program.

The stagnation streamline approach may be inappropriate for flows with thermal radiation in that electronically excited particles may be removed before spontaneous emission occurs (Moss et al. 1988).

The test case was for a hypersonic flow of nitrogen with the removal region specified such that the complete stagnation streamline was modelled. The surface temperature was small in comparison with the temperature behind the shock and this led to a very large density increase in a narrow region adjacent to the surface. The cell structure of the other programs is inadequate under these circumstances. The flowfield has therefore been divided into a number of regions and each of these has a separate value of DTM and FNUM, although the ratio of these quantities is the same in every region. Because the ratio is the same, the flux of molecules between the regions is matched and there is no need for any removal or duplication of molecules crossing from one region to another. This technique can be used only for flows that become steady, and the flowfield is realistic only when steady flow has been attained. While this example involves a density variation of only two orders of magnitude, the technique makes possible the efficient simulation of flows with density variations of many orders of magnitude.

The stream speed was $-10{,}000\ \mathrm{m\,s^{-1}}$ and the freestream number density and temperature were $10^{20}\ \mathrm{m^{-3}}$ and 180 K, respectively. The molecule removal region extended from the origin at $x=0$ to $x=0.1$, while the boundary of the simulated region was at $x=0.6$. The resulting velocity profile is shown in Fig. 12.45 and it can be seen that the molecule removal commences near the downstream end of the shock wave.

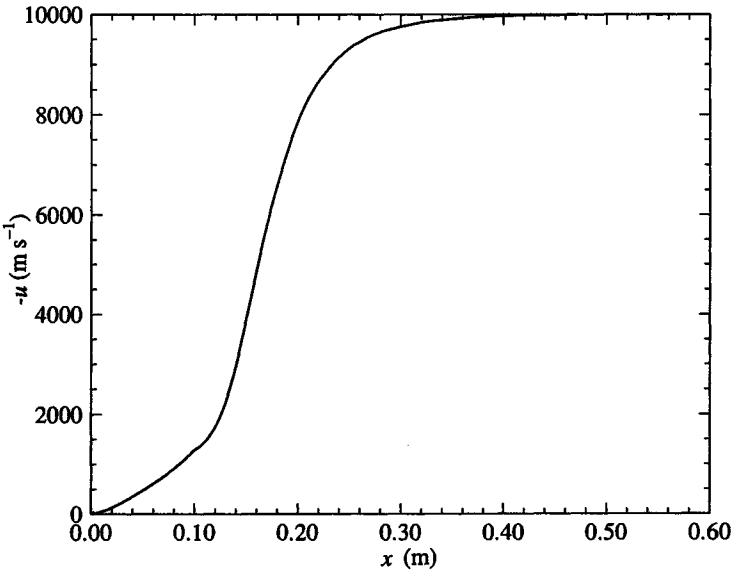


Fig. 12.45 The velocity profile along the stagnation streamline.

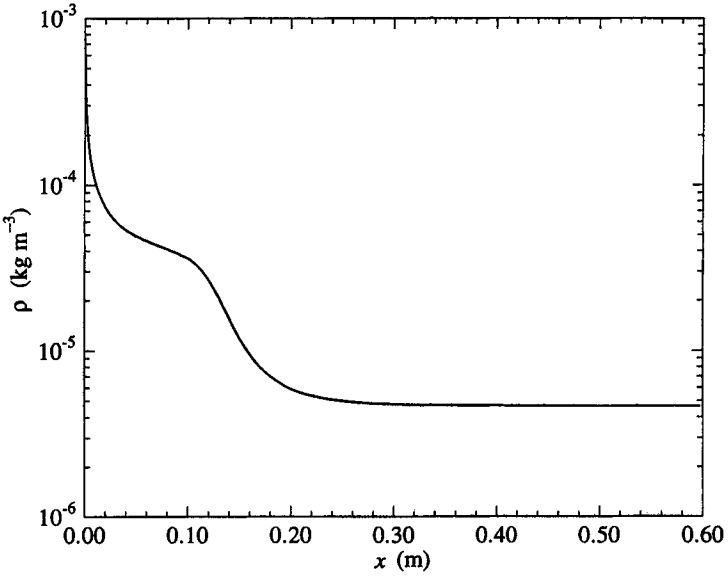


Fig. 12.46 The density rises across the shock and boundary layer.

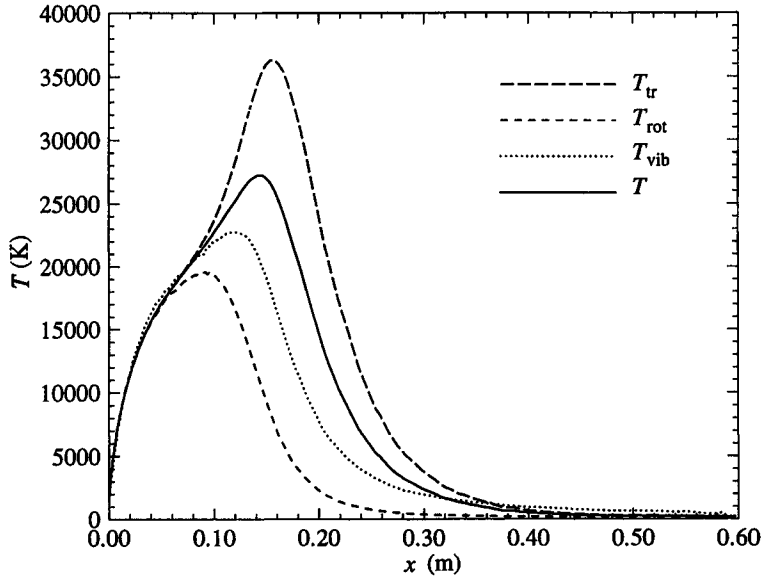


Fig. 12.47 The components of temperature along the streamline.

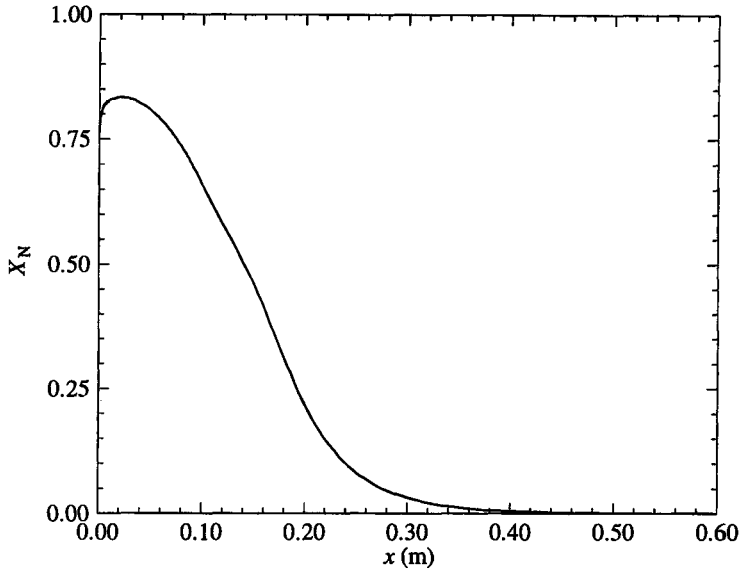


Fig. 12.48 The fraction of atomic nitrogen along the streamline.

There is a factor of ten increase in density across the shock wave and, as shown in Fig. 12.46, a further factor of ten increase across the thermal boundary layer to the surface at 1000 K. The density gradient in the thermal boundary layer is actually higher than that in the shock wave, but Fig. 12.47 shows that the separate rotational, vibrational, and translational temperatures are in equilibrium in the boundary layer. The initial temperature rise occurs well upstream of any significant increase in density, and there is extreme non-equilibrium between the separate components of the temperature in the upstream half of the wave. The temperatures are sufficiently high to cause the dissociation of a significant fraction of the nitrogen, and the number fraction of atomic nitrogen is shown in Fig. 12.48. The fall in the concentration of atomic nitrogen near the surface is a consequence of thermal diffusion.

The near linear decrease in the stream speed in the molecule removal to zero at the stagnation point is similar to that in blunt body flows, and the surface properties are expected to be close to those in axially symmetric or two-dimensional flows that have the same shock standoff distance. The net heat transfer to the surface for the steady state of this flow was 0.28 MWm^{-2}, and this is just over one tenth of the energy flux in the free-stream. The heat transfer due to the translational and internal energies was 0.43 and 0.05 MWm^{-2}, respectively, while that due to the reflected molecules was −0.18 and −0.02 MWm^{-2}. The pressure due to the incident molecules was 246 Nm^{-2} and 163 Nm^{-2} due to the reflected molecules.

12.13 Adiabatic atmosphere

A gravitational field can be included in applications of program DSMC1 to plane flows. A case was set up with a stationary gas of argon with number density 10^{24} m^{-3} and a temperature of 200 K at the inner boundary at $x=0$. The outer boundary at $x=1$ m was a vacuum and the initial state was a uniform stationary gas at the inner boundary conditions. The continuum solution for a static adiabatic atmosphere predicts a constant temperature lapse rate of

$$\frac{dT}{dx} = \frac{\gamma-1}{\gamma}\frac{mg}{k}. \tag{12.39}$$

The 'gravitational' acceleration was set to -1.25×10^5 m/s^2 and the resulting lapse rate of just over 240 Km^{-1} is such that the 'atmosphere' should terminate at about 0.83 m. The unsteady phase of the flow involved strong disturbances as the gas expands into the vacuum while being restrained by the gravitational field. These disturbances are slow to decay, but the eventual establishment of a steady flow is verified in Fig. 12.49 by plotting the time-averaged temperature profile over three separate time intervals. The temperature follows the continuum prediction for a short distance, but then falls at a much slower rate and the atmosphere extends out to the vacuum boundary.

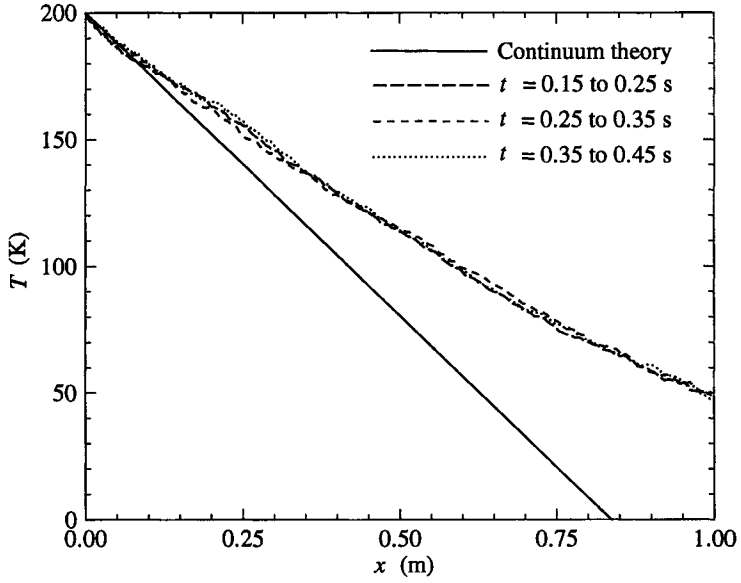


Fig. 12.49 The temperature profile in the adiabatic atmosphere.

The local Knudsen number, based on the density scale length, at the point where the temperature departs from the continuum value is approximately 0.002. The pressure gradient is everywhere consistent with the continuum equation

$$dp/dx = \rho g . \tag{12.40}$$

Note that in both these equations and in program DSMC1 g is positive when it is directed in the positive x-direction. The density at the outer boundary is approximately 1% of the density of the stationary gas that constitutes the inner boundary. The atmosphere is not stationary and the velocity ranges from approximately 0.7 ms^{-1} near $x = 0$ to 70 ms^{-1} at the outer boundary.

The calculation was repeated for a mixture of equal parts by number of argon and helium. The gravitational field was adjusted so that the continuum lapse rate was the same as in the simple gas case. As shown in Fig. 12.50 there was no separation of these gases in the atmosphere. This result is perhaps surprising, but it should be noted that the variable $\boldsymbol{F}$ in both the Boltzmann equation and the diffusion equation is a force per unit mass and does not lead to separation effects in a gas mixture. Any separation would be due to either pressure or thermal diffusion. Thermal diffusion would act to concentrate the argon at the outer boundary and pressure diffusion would tend to concentrate it at the inner boundary. It appears that the cancellation of these effects is almost exact.

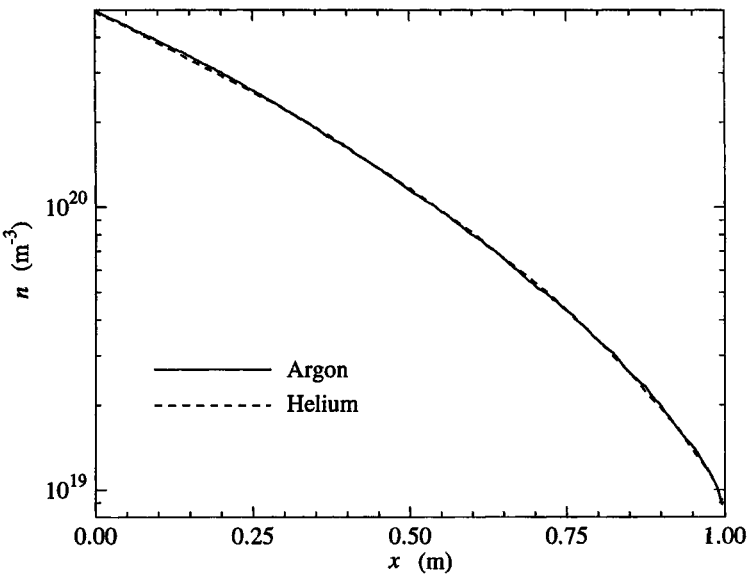


Fig. 12.50 Number density profiles in a mixture of equal parts of argon and helium.

12.14 Gas centrifuge

Consider a cylindrical flow with the axis as the inner boundary and a surface with a circumferential velocity of 1000 m/s as the outer boundary. The gas is initially a mixture of equal parts by number argon and helium that is stationary at a number density of 10^{21} m^{-3} and a temperature of 200 K. The outer moving boundary is diffusely reflecting at a temperature of 200 K. The rotation of the outer boundary effectively starts at zero time and a steady flow is eventually established.

Unlike the preceding case, there is no force acting directly on the molecules but, as shown in Fig. 12.51 for the number density profiles of the separate species, the separation effects are significant. The circumferential velocity and the temperature profiles are shown in Figs. 12.52 and 12.53, respectively. These profiles show that pressure and thermal diffusion act in the same direction in the outer part of the flow, but the pressure diffusion would be dominant. Adjacent to the surface, the argon number density is double the helium number, but the region of enhanced argon extends into the flow only to 0.93 m. There is then a region of enhanced helium that extends to the radius of approximately 0.5 m. The samples are so small near the axis that the apparent separation in this region is probably not significant.

The circumferential velocity declines to near zero near a radius of 0.5 m and the conditions in the inner core of the flow are not significantly different from the initial conditions.

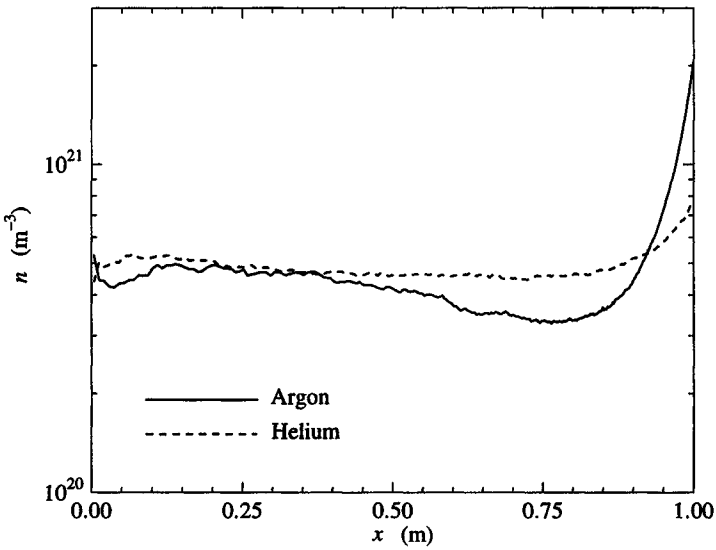


Fig. 12.51 The number density profiles in the gas centrifuge.

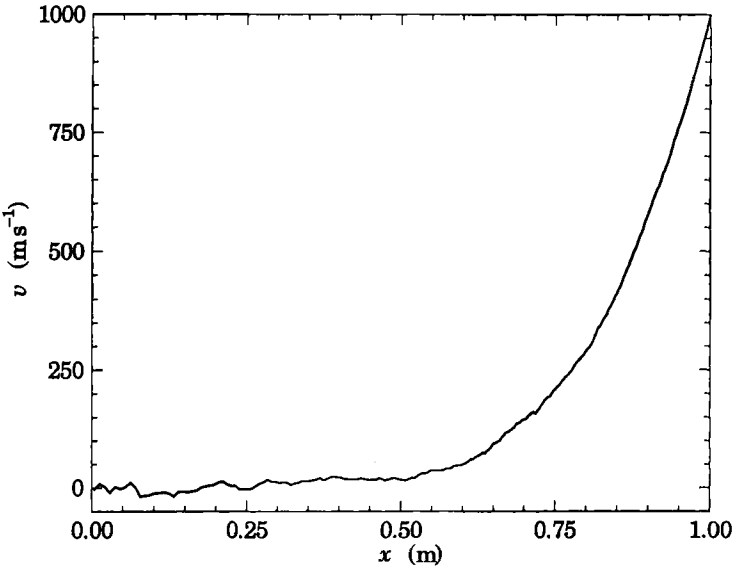


Fig. 12.52 The circumferential velocity profile in the centrifuge.

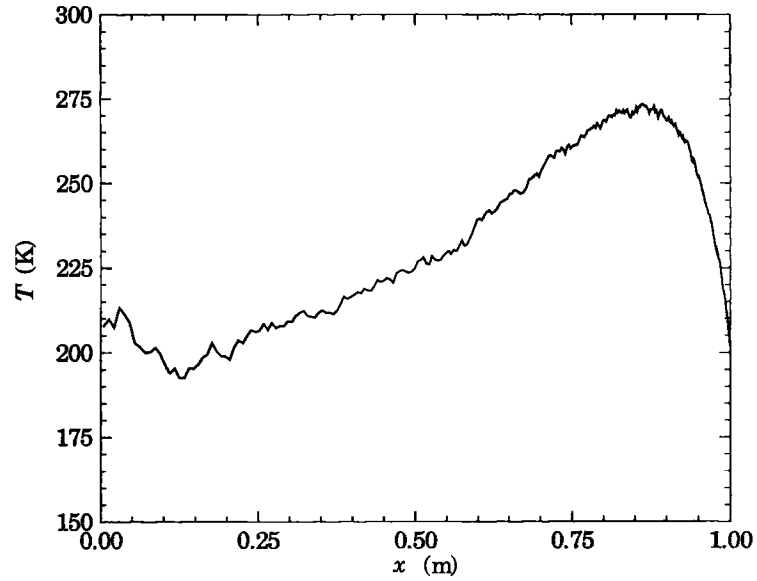


Fig. 12.53 The temperature profile in the centrifuge.

The temperature of the gas adjacent to the surface is only slightly above the surface temperature of 200 K, but it increases to a maximum of just over 270 K at a radius of 0.87 m. This means that there is a net heat transfer to the surface, and this equals the work done against the shear stress by the rotation of the cylinder.

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